# TOPS NETWORK SEGMENTATION

### Cisco Packet Tracer — Practical Documentation

Objective: Configure VLAN 10 Students, VLAN 20 Faculty, VLAN 30 Admin, router-on-a-stick routing, access-control rules, and ping verification.

## 1. Required permissions

| From → To | Students VLAN 10 | Faculty VLAN 20 | Admin VLAN 30 |
|---|---|---|---|
| Students (10) | Allow | Deny | Deny |
| Faculty (20) | Allow | Allow | Deny |
| Admin (30) | Deny | Allow | Allow |

## 2. Topology and cables

- Router0 Fa0/0 ↔ Switch0 Fa3/1 (trunk)
- Student PC0 ↔ Switch0 Fa2/1 (access VLAN 10)
- Faculty PC1 ↔ Switch0 Fa0/1 (access VLAN 20)
- Admin PC2 ↔ Switch0 Fa1/1 (access VLAN 30)

Use Copper Straight-Through or Auto-Connect. If your actual ports differ, substitute the interface names consistently.

## 3. Configure Switch0

Open Switch0 → CLI. If asked to enter the initial configuration dialog, type no.

```
enable
configure terminal
vlan 10
 name STUDENTS
exit
vlan 20
 name FACULTY
exit
vlan 30
 name ADMIN
exit
interface fastEthernet 2/1
 switchport mode access
 switchport access vlan 10
 spanning-tree portfast
 no shutdown
exit
interface fastEthernet 0/1
 switchport mode access
 switchport access vlan 20
 spanning-tree portfast
 no shutdown
exit
interface fastEthernet 1/1
 switchport mode access
 switchport access vlan 30
 spanning-tree portfast
```

```
 no shutdown
exit
interface fastEthernet 3/1
 switchport mode trunk
 switchport trunk allowed vlan 10,20,30
 no shutdown
exit
end
write memory
```

## 4. Configure Router0 — router-on-a-stick

```
enable
configure terminal
interface fastEthernet 0/0
 no shutdown
exit
interface fastEthernet 0/0.10
 encapsulation dot1Q 10
 ip address 192.168.10.1 255.255.255.0
exit
interface fastEthernet 0/0.20
 encapsulation dot1Q 20
 ip address 192.168.20.1 255.255.255.0
exit
interface fastEthernet 0/0.30
 encapsulation dot1Q 30
 ip address 192.168.30.1 255.255.255.0
exit
```

## 5. Configure ACLs for the requested ping policy

These ACL rules implement the directional ICMP ping tests. Echo-reply exceptions allow permitted pings to receive replies. This is a ping-focused lab policy, not a stateful firewall for every application.

```
ip access-list extended STUDENTS-IN
 permit icmp 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255 echo-reply
 deny ip 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255
 deny ip 192.168.10.0 0.0.0.255 192.168.30.0 0.0.0.255
 permit ip any any
exit
ip access-list extended FACULTY-IN
 permit icmp 192.168.20.0 0.0.0.255 192.168.10.0 0.0.0.255 echo
 permit icmp 192.168.20.0 0.0.0.255 192.168.30.0 0.0.0.255 echo-reply
 deny ip 192.168.20.0 0.0.0.255 192.168.30.0 0.0.0.255
 permit ip any any
exit
ip access-list extended ADMIN-IN
 permit icmp 192.168.30.0 0.0.0.255 192.168.20.0 0.0.0.255 echo
 deny ip 192.168.30.0 0.0.0.255 192.168.10.0 0.0.0.255
 permit ip any any
exit
interface fastEthernet 0/0.10
```

```
 ip access-group STUDENTS-IN in
exit
interface fastEthernet 0/0.20
 ip access-group FACULTY-IN in
exit
interface fastEthernet 0/0.30
 ip access-group ADMIN-IN in
exit
end
write memory
```

## 6. Configure PC IP addresses

Each PC: Desktop → IP Configuration.

| PC / VLAN | IPv4 address | Subnet mask | Default gateway |
|---|---|---|---|
| Student PC0 / VLAN 10 | 192.168.10.10 | 255.255.255.0 | 192.168.10.1 |
| Faculty PC1 / VLAN 20 | 192.168.20.10 | 255.255.255.0 | 192.168.20.1 |
| Admin PC2 / VLAN 30 | 192.168.30.10 | 255.255.255.0 | 192.168.30.1 |

## 7. Verify configurations

On Switch0, use privileged mode (Switch#):

```
show vlan brief
show interfaces trunk
show running-config
```

On Router0, use privileged mode (Router#):

```
show ip interface brief
show ip route
show access-lists
show running-config
```

If the prompt shows (config)#, type end before running show commands.

## 8. Ping tests

| Run from | Command (Desktop → Command Prompt) | Expected result |
|---|---|---|
| Student PC0 | ping 192.168.10.1 | Allow — gateway |
| Student PC0 | ping 192.168.20.10 | Deny |
| Student PC0 | ping 192.168.30.10 | Deny |
| Faculty PC1 | ping 192.168.10.10 | Allow |
| Faculty PC1 | ping 192.168.20.1 | Allow — gateway |
| Faculty PC1 | ping 192.168.30.10 | Deny |
| Admin PC2 | ping 192.168.20.10 | Allow |
| Admin PC2 | ping 192.168.10.10 | Deny |
| Admin PC2 | ping 192.168.30.1 | Allow — gateway |

The first ping may time out while ARP resolves; repeat the ping once before deciding it failed.

## 9. Troubleshooting

- Confirm PC IP, subnet mask, and gateway.
- Confirm each PC port is assigned to the correct VLAN.
- Confirm Switch0 Fa3/1 is trunking and allows VLANs 10,20,30.
- Confirm Router0 Fa0/0 and subinterfaces Fa0/0.10, Fa0/0.20, Fa0/0.30 are up/up.
- If your router uses GigabitEthernet, change the interface names to match the actual device.